MIL-P-60601B (AR)
16 March 1981

SUPERSEDING
MIL-P-60601A (MU)
3 June 1975

MILITARY SPECIFICATION

PROJECTILE, 105MM, WP, M416
METAL PARTS ASSEMBLY

This specification is approved for use by the US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, 105MM, WP, M416 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-6867 - Magnetic Inspection Units
MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1169- Packaging, Packing and Marking of Inert Ammunition Components

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND
8837835 - Pallet for Fixed and Semifixed Shells
10522803 - Projectile, 105MM, WP, M416 Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officers.)

2.2 Other Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI - B46.1 Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018.)

ASTM B117 - Salt Spray (Fog) Testing
ASTM E8 - Tension Testing of Metallic Materials
ASTM E18 - Tests for Rockwell Hardness and Superficial Hardness for Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply

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with the requirements specified on the applicable drawings and referenced specifications.

3.2 Heat Treatment or Stress Relief.- Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce the mechanical properties specified on the applicable drawing.

3.2.1 Hot Forge - Heat Treat Method.- Prior to final machining, projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment applied shall consist of heating the projectile bodies above the critical temperature, not exceeding 1600 degrees Fahrenheit (^{o}F), (871 degrees Celsius (^{o}C)), followed by an oil quench. After quenching the body forgings shall be tempered without delay at a suitable temperature above 750°F (389.9^{o}C) to produce the required mechanical properties. After removal from tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700°F (371.1^{o}C), after which cooling may be accelerated by water spray.

3.2.2 Cold shaping and hot cup - Cold Draw Method.-Cold working shall be accomplished as to produce the mechanical properties specified in the projectile body. The projectile body shall be stress relieved at a temperature of 700°F (371.1^{o}C), minimum, for a period of thirty minutes, minimum, following the final forming operation during which time plastic flow takes place.

3.3 Mechanical properties

3.3.1 Body and base plug.-The mechanical properties specified by the applicable drawings shall apply throughout each component, when tested as specified in 4.4.2.2 and 4.4.2.3.

3.3.2 Rotating bands.-After application and subsequent removal from the projectile, the rotating bands shall have the following mechanical properties, when tested as specified in 4.4.2.4.

Tensile Strength - 32,000 pounds per square inch (psi)
2.2064 Mega Pascals (MPA), minimum

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Elongation - 15 percent per inch (2.54 centimeter (cm)) of gage length

3.4 Banding

3.4.1 Band seat.-Prior to the assembly of rotating band blanks, the band seat shall be clean and free from oil, grease,dirt, rust or other foreign matter. Shot or sand blasting shall not be permitted for cleaning the band seats following knurling.

3.4.2 Band blanks.-The rotating band blanks shall be in the form of a solid ring, annealed, if necessary to comply with the requirements of 8597101 and the applicable specification.

3.4.3 - Application.- The rotating band blanks shall be applied after final heat treatment or stress relief of the projectile body by carefully pressing them into the band seats so that contact is made with the sidewalls for eight-tenths of the band seat depth, minimum. There shall be no gap between the band and sidewall at the outside surfaces of the body after the final machining of the rotating bands. During banding, the complete inside diameter of the rear of the projectile body under both band seats shall be supported to prevent deformation of the projectile body.

3.4.4 Band Seating.- The diametrical clearance between the rotating band and band seat shall not exceed six-thousandths of an inch (0.006 inches), when tested as specified in 4.5.5.1.

3.5 Defective Raw Material and Metal Parts Components

3.5.1 Billet.- The billet shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, folds, bursts, seams, and other metal defects.

3.5.2 Projectile Body and Press Fit Surface.- The projectile body, including the press fit surface, shall be free from cracks, seams, splits, stringers, laps, folds, slugs, porosity, dents, scale, fins, internal burrs, draw marks, laminations, pits, sharp edges and imbedded foreign matter.

3.5.3 Base Plug.- The base plug shall be free from cracks, folds or other metal defects.

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3.5.4 Visual Standards.- Visual standards depicting maximum acceptable degrees of conditions described in 3.5.1, 3.5.2 and 3.5.3 will be established for each contract after the first month's or lot's production (whichever quantity is smaller) or as specified in the contract. In the absence of a visual standard, the requirements of 3.5.1., 3.5.2 and 3.5.3 shall apply (See 4.4.2-C).

3.6 Proving Ground Tests.- The assembled projectile metal parts shall be safe to fire and shall reveal no evidence of failure when tested as specified in 4.5.7. Metal parts failure shall be:

a. Loss or break-up of metal parts components during gun launch or flight.

b. Loosening or fanning of the rotating band(s).

c. Shearing of the rotating band(s) as evidence by a complete lack of engraving.

d. Leakage of propellant gases into the base plug thread area.

3.7 Surface Preparation and Application of Projectile Finish.- Where cleaning and surface pretreatment are required by component details, the process shall apply to all indicated surfaces regardless of contour configuration. The protective finish shall be as specified on the applicable drawings and shall be uniformly applied to the surfaces designated by the applicable drawings. Salt spray and other required testing shall be in accordance with the applicable drawings and specifications.

3.8 Data Requirements.- The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.4.2.b.

3.9 First Article Inspection.- This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship.- The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following.

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3.10.1 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.10.2 Burr.- No part shall have a burr.

3.10.3 Foreign Matter.- All parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

3.10.4 Rotating Band Damage.- The rotating band after final machining shall be protected from being nicked, burred or otherwise damaged during subsequent handling and preparation for shipment.

3.10.5 Painting and Marking.- Painting and marking shall be applied in accordance with the applicable drawings and specifications. The rotating bands and thread of the projectile body shall be protected to prevent the application of paint to these surfaces.

4. QUALITY ASSURANCE

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.- The following types of inspection shall be performed on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.- The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

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Part Description	Drawing	Quantity
Gasket	8597096	5
Base Plug	10522806	25
Body	10522805	25
Body assembly, prior to painting	10522804	5
Body assembly, after painting	10522803	5
Projectile, 105MM, WP, M416		
Metal Parts Assembly	10522803	20

4.3.2 Inspections to be Performed.- See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.- See MIL-A-48078

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TABLE I - FIRST ARTICLE

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
	Projectile, 105MM, WP, M416 Metal Parts		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Gasket (Dwg. 8597096)				
	Examination for Defects	5		3.1	4.4.2.1
	Hardness	5		3.1	
	Plug, Base (Dwg. 10522806)				
	Examination for Defects	25		3.1	4.4.2.2
	Mechanical Properties	3		3.3	
	Salt Spray Resistance	2		3.7	
	Body (Dwg. 10522805)				
	Examination for Defects	25		3.1	4.4.2.3
	Hardness	25		3.1	
	Mechanical Properties	3		3.3	
	Body Assembly, Prior to Painting (Dwg. 10522804)				
	Examination for Defects	5		3.1	4.4.2.4
	Rotating Band Mechanical Properties	2		3.3.2	
	Rotating Band Seating	2		3.4	
	Rotating Band Sidewall Contact	2		3.4	

NOTES:

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TABLE I - FIRST ARTICLE

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	CONTINUATION SHEET				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body Assembly, After Painting (Dwg. 10522803)				
	Examination for Defects	5		3.1	4.4.2.5, 4.4.2.6
	Salt Spray Resistance	2		3.7	
	Projectile, 105MM, WP, M416 Metal Parts Ballistic Test				
	Phase				
	Security	10		3.6	4.5.7.1
	Performance	10		3.6	4.5.7.2

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.- Inspection lots shall comply with the lot formation provision of MIL-A-48078.

4.4.1.1 Heat Treatment Lots.- Heat treatment lots shall contain body forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test prescribed herein.

4.4.1.2 Stress Relieved Lots.- Stress relieved lots shall contain cold worked bodies from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment, and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test prescribed herein.

4.4.1.3 Tempered Lots.- Tempered lots shall contain base plugs from not more than one heat of aluminum alloy that have been formed by the same process, utilizing the same tooling and equipment, and tempered and aged in the same furnaces at the same temperature for the same length of time. The tempered lot thus constituted shall be the basis for the selection of tensile samples as prescribed herein.

4.4.1.4 Projectile Metal Parts Assembly Lots - Inspection lots of projectile metal parts assemblies shall contain projectile bodies from not more than two heats of steel and base plugs from not more than two heats of aluminum produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision under condition of continuous production free from interruptions other than those due to the end of a shift, day or workweek. The projectile metal parts assembly lot thus constituted shall be the basis for the selection of the sample for the proving ground and functional tests prescribed herein.

4.4.2 Examination

a. Sampling Plans.- Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Inspection Level II.

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b. Defect Reporting.- The contractor shall prepare a monthly report of all critical defects found, including the number of projectile metal parts assemblies inspected and any corrective action taken, in accordance with Data Item Description DI-R-1721, and deliver them as specified in the contract.

c. Visual Inspections.- In the event of a dispute, four power (4x) magnification shall be used in performing inspections for defects such as burrs and foreign matter.

d. Surface Finish.- The roughness comparison specimens or methods to determine the average surface roughness prescribed by ANSI-B46.1 may be used as a basis for the surface finish determination of a component.

e. Control Tool Manufacture.- Dimensions marked tool control shall be gaged at the beginning of each eight hours production or fraction thereof from each machine in operation; and, prior to any tooling change. Where the destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted, provided that the contractor has established a correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. All material produced prior to periodic inspection shall be rejected whenever a defective tool control dimension is found.

f. Inspection of Slug Prior to Forming.- Each slug obtained from a billet or bloom shall be visually inspected for surface indications of cracks, splits, pipe, porosity, folds, seams and other metal defects visible to the unaided eye on all exterior surfaces of the slug. Visual standards may be selected to aid in the determinaton of a slugs acceptability.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8597096
4.4.2.1	Gasket				NEXT HIGHER ASSEMBLY 10522803
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	True position of outside diameter with inside diameter		0.40%	3.1	Gage
102	Inside diameter		0.40%	3.1	Gage
103	Outside diameter		0.40%	3.1	Gage
104	Thickness		0.40%	3.1	Gage
105	Hardness		0.40%	3.1	Gage
MINOR					
201	Surface Finish Improper		0.65%	3.1	Gage
202	Presence of a Burr		0.65%	3.10	Visual/Manual
203	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.2	Plug, Base				10522806
					NEXT HIGHER ASSEMBLY
					10522803
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal defective		100%	3.5	Visual
MAJOR					
101	Runout of pitch diameter of internal thread with pitch diameter of external thread, including front face and bottom of internal thread rear counterbore		0.40%	3.1	Gage
102	True position of rear counterbore with pitch diameter of internal thread		0.40%	3.1	Gage
103	Pitch diameter of external thread, min.		0.40%	3.1	Gage
104	Major diameter of external thread, min.		0.40%	3.1	Gage
105	Pitch diameter of internal thread, max.		0.40%	3.1	Gage
106	Minor diameter of internal thread, max.		0.40%	3.1	Gage
107	Outside diameter forward of external thread		0.40%	3.1	Gage
108	Diameter of forward counterbore		0.40%	3.1	Gage
109	Diameter of rear counterbore		0.40%	3.1	Gage
110	Diameter of dovetail		0.40%	3.1	Gage
111	Outside diameter rear of external thread		0.40%	3.1	Gage
112	Depth of forward counterbore		0.40%	3.1	Gage
113	Depth of rear counterbore		0.40%	3.1	Gage
114	Length from front face to start of external thread		0.40%	3.1	Gage
115	Length from front face to rear of dovetail section		0.40%	3.1	Gage

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	CONTINUATION SHEET				10522806
					NEXT HIGHER ASSEMBLY 10222803
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Length of dovetail		0.40%	3.1	Gage
117	Length, total		0.40%	3.1	Gage
118	Thread(s) damaged or not full	See	0.40%	3.1	Visual
119	Mechanical properties	4.4.3.1		3.3	4.5.1
120	Salt spray resistance test	See 4.4.3.3		3.7	4.5.6
MINOR					
201	Pitch diameter of external thread, max.		0.65%	3.1	Gage
202	Pitch diameter of internal thread, min.		0.65%	3.1	Gage
203	Location of wrench slots		0.65%	3.1	Gage
204	Width of wrench slots		0.65%	3.1	Gage
205	Depth of wrench slots		0.65%	3.1	Gage
206	Surface finish improper		0.65%	3.1	Gage
207	Radii or chamfers missing or incorrect		0.65%	3.1	Visual/Gage
208	Presence of a burr		0.65%	3.10	Visual/Manual
209	Protective coating inadequate or damaged		0.65%	3.7	Visual
210	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.3	Body				10422805
					NEXT HIGHER ASSEMBLY
					10522804
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal defective		100%	3.5	Visual
2	Magnetic particle inspection		100%	3.5	4.5.2
MAJOR					
101	Runout of pitch diameter of thread with bourrelet and base plug seat		0.40%	3.1	Gage
102	True position of dovetail with pitch diameter of thread		0.40%	3.1	Gage
103	Pitch diameter of thread, max.		0.40%	3.1	Gage
104	Minor diameter of thread, max.		0.40%	3.1	Gage
105	Diameter at basic length forward of press fit surface		0.40%	3.1	Gage
106	Depth of thread cavity		0.40%	3.1	Gage
107	Thickness of wall at nose		0.40%	3.1	Gage
108	Thickness of wall in ogive		0.40%	3.1	Gage
109	Thickness of wall at forward basic length		0.40%	3.1	Gage
110	Variation in wall thickness (for indicated length)		0.40%	3.1	Gage
111	Diameter of forward rotating band seat		0.40%	3.1	Gage
112	Diameter of rear rotating band seat		0.40%	3.1	Gage
113	Width of forward rotating band seat		0.40%	3.1	Gage
114	Width of rear rotating band seat		0.40%	3.1	Gage

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.3	CONTINUATION SHEET				10522805
					NEXT HIGHER ASSEMBLY
					10522804
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115	Length of effective thread, min.		0.40%	3.1	Gage
116	Thread damaged or not full		0.40%	3.1	Visual
117	Hardness		100%	3.1	4.5.1/Test
118	Mechanical properties	See			4.5.1/Test
119	Runout of ogive	4.4.3.1	0.40%	3.3	Gage
MINOR					
201	Pitch diameter of thread, min.		0.65%	3.1	Gage
202	Diameter of thread chamfer		0.65%	3.1	Gage
203	Diameter at rear of dovetail		0.65%	3.1	Gage
204	Depth of dovetail section		0.65%	3.1	Gage
205	Length, total		0.65%	3.1	Gage
206	Surface finish improper		0.65%	3.1	Gage
207	Location of forward rotating band seat		0.65%	3.1	Gage
208	Location of rear rotating band seat		0.65%	3.1	Gage
209	Scoring missing from rotating band seat ribs		0.65%	3.1	Visual
210	Foreign matter in band seat		0.65%	3.1	Visual
211	Radii or chamfers missing or incorrect		0.65%	3.1	Visual/Gage
212	Presence of a burr		0.65%	3.10	Visual/Manual
213	Evidence of poor workmanship		0.65%	3.10	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.4	Body Assembly, Prior to Painting		10522804
			NEXT HIGHER ASSEMBLY
			10522803

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal defective			3.5	Visual
2	Diameter of press fit surface			3.1	Gage
3	Reverse taper of press fit surface			3.1	Gage
4	Length from rear face to front of press fit surface, min.			3.1	Gage
5	Radius not tangent to press fit surface for full 360 degree rotation			3.1	Visual/Gage
6	Hydrostatic pressure test			3.1	4.5.3
7	Air pressure test (if applicable)			3.1	4.5.4
MAJOR					
101	Profile of nose (after hydrostatic test)		100%	3.1	Gage
102	Runout of press fit surface with bourrelet		0.40%	3.1	Gage
103	Runout of forward rotating band with bourrelet		0.40%	3.1	Gage
104	Runout of rear rotating band with bourrelet		0.40%	3.1	Gage
105	Diameter of bourrelet, min.		0.40%	3.1	Gage
106	Diameter forward of forward rotating band, min.		0.40%	3.1	Gage
107	Diameter between rotating bands, min.		0.40%	3.1	Gage
108	Diameter rear of rear rotating band, min.		0.40%	3.1	Gage
109	Location of forward rotating band		0.40%	3.1	Gage
110	Location of rear rotating band		0.40%	3.1	Gage
111	Diameter of forward rotating band		0.40%	3.1	Gage
112	Diameter of rear rotating band		0.40%	3.1	Gage

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PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.4	CONTINUATION SHEET				10522804
					NEXT HIGHER ASSEMBLY
					10522803
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Rotating band mechanical properties	See 4.4.3.1.3		3.3.2	4.5.1
114	Rotating band seating	See 4.4.3.2		3.4.1	4.5.5
115	Rotating band sidewall contact	See 4.4.3.2		3.4.3	4.5.5
MINOR					
201	Width of forward rotating band		0.65%	3.1	Gage
202	Width of rear rotating band		0.65%	3.1	Gage
203	Depth of undercut front and rear of forward rotating band (if applicable)		0.65%	3.1	Gage
204	Depth of undercut front and rear of rear rotating band (if applicable)		0.65%	3.1	Gage
205	Width of undercut front and rear of forward rotating band (if applicable)		0.65%	3.1	Gage
206	Width of undercut front and rear of rear rotating band (if applicable)		0.65%	3.1	Gage
207	Radii or chamfers missing or incorrect		0.65%	3.1	Visual/Gage
208	Surface finish of bands improper		0.65%	3.1	Gage
209	Presence of a burr		0.65%	3.10	Visual/Manual
210	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Body Assembly, After Painting				10522803
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal defective		100%	3.5	Visual
2	Surface finish of press fit surface improper		100%	3.1	Gage
3	Any inspection acceptance stamp missing		100%	3.1	Visual
4	Any inspection rejection stamp present		100%	3.1	Visual
MAJOR					
101	Diameter forward of rotating band, max.		100%	3.1	Gage
102	Weight		100%	3.4	Approved Scale
103	Salt spray resistance	See 4.4.3.3		3.7	Test
MINOR					
201	Paint on thread(s)		0.65%	3.10	Visual
202	Paint on rotating band		0.65%	3.10	Visual
203	Rotating band nicked or damaged		0.65%	3.10	Visual/Manual
204	Foreign matter present		0.65%	3.10	Visual
205	Protective coating inadequate or damaged		0.65%	3.10	Visual
206	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Projectile, WP, 105MM, M416 Metal Parts Assembly				10522803
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal defective			3.7	Visual
2	Ballistic test - security phase	10		3.6	4.5.7.1
MAJOR					
101	Corrosion preventive compound missing from threads or press fit surface area		0.40%	3.1	Visual
102	VCI tablets or crystals missing		0.40%	3.1	Visual
103	Ballistic test - performance phase	10		3.6	4.5.7.2
MINOR					
201	Protective cap missing		0.65%	3.1	Visual
202	Marking missing, incorrect or illegible		0.65%	3.10	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Carton, Packed (Base Plug) Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Support missing or damaged		0.65%	3.1	Visual
202	Separator(s) missing or damaged		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Carton, Packed (Base Plug) After Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Contents loose		0.65%	3.1	Manual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton improperly sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60601B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Pallet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Strapping missing, broken or loose		0.65%	3.1	Visual/Manual
202	Pallet damaged		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Mechanical Properties

4.4.3.1.1 Projectile Body Hardness

4.4.3.1.1.1 Heat Treatment Lots - In order to determine the uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.4.1.1) shall be subjected to a hardness test as prescribed in 4.5.1.1.

4.4.3.1.1.2 Stress Relieved Lots - Twenty-five bodies from the first batch of each stress relieved lot (as defined by 4.4.1.2) shall be subjected to a hardness test as prescribed in 4.5.1.1. Following satisfactory compliance of the results of the hardness test for the sample, five projectile bodies from each succeeding batch of a stress-relieved lot shall be subjected to this test.

4.4.3.1.1.2.1 Control Chart for Hardness - A control chart for projectile body hardness shall be constructed for each stress relieved lot. The hardness values obtained from the sample of twenty five bodies from the first batch of each stress relieved lot, if acceptable in accordance with the requirements of the applicable drawing, shall be used to construct these control charts using ninety nine percent (99%) boundary lines for the remainder of the stress relieved lot. If, in any succeeding batch, any individual reading on any projectile body in the sample of five bodies exceeds the boundary lines calculated for that stress relieved lot, a second sample selected in accordance with MIL-STD-105, Inspection Level I and an AQL of 0.40% shall be taken from that batch. If the average hardness of each projectile body in this second sample complies with the control chart requirements, then the batch shall be accepted and sampling shall resume at the five sample rate for each successive batch in the remainder of the stress relieved lot. If any average hardness reading falls outside the control chart constraints, the batch shall be rejected and a new stress relieved lot shall be established in accordance with 4.4.3.1.1.2.

4.4.3.1.2 Projectile Body Tensile Test

4.4.3.1.2.1 Heat Treatment Lots - Two projectile bodies, one of the hardest and one of the softness, as determined by

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the hardness test, shall be subjected to this test. Failure of any tensile test specimen obtained from these samples to comply with the yield strength and percent elongation requirements of the applicable drawing shall reject the heat treatment represented by the tensile samples. At the option of the contractor and only with the prior approval of the Contracting Officer, a production control sample may be utilized to establish a hardness range which will produce acceptable tensile tests of projectile body samples selected from both extremes of the hardness range. When such a control sample is used and a hardness range thus established for a heat treatment lot, the hardness range shall remain the same throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies whose hardness fall outside of the established range may be re-heattreated to fall within the range; or, the mechanical properties may be determined from another sample of projectile bodies and a new heat treatment lot established.

4.4.3.1.2.2 Stress Relieved Lots - Two projectile bodies determined to be the hardest and the softest by the hardness test of the twenty five samples shall be subjected to this test. Failure of any of the tensile test samples obtained from these bodies shall reject the stress relieved lot represented by the samples.

4.4.3.1.2.2.1 Stress Relieving - When it has been determined that the mechanical properties of the projectile body samples selected from the stress relieving lot comply with the requirements of the applicable drawing, the furnace temperature and cycle (soak) time established for the stress relieved lot shall be maintained throughout the lot. The furnace temperatures may vary within 25°F (16.1 °C) of the temperature established; but, shall not be allowed to fall below 700°F (371.1 °C) for the thirty minute cycle time, minimum. The furnace temperature and elapsed time shall be recorded at least once every hour to assure adequate control of the contractor's stress relieving process and the furnace temperature shall be calibrated at least once every eight hours of operation by an independent method.

4.4.3.1.3 Rotating Band Tensile Strength - For each shifts production or fraction thereof, one projectile body assembly from each banding machine in operation shall be subjected to this test as described in 4.5.1.2.2. Failure of either of the

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two bands from the sample body assembly to comply with the requirements of 3.3.2, shall result in rejection of the quantity represented by the sample.

4.4.3.1.4 Tensile Strength of Base Plug - Two base plugs from the first batch of a stress relieved lot shall be tested to determine compliance with the yield and percent elongation requirements of the applicable drawing in order to qualify the stress relieving process for that lot. Following the establishment of a stress relieved lot, two base plugs from every five batches thereafter in the lot shall be subjected to this test. Failure of any base plug tensile specimen to comply with the applicable requirements shall reject the batch or lot of base plugs represented by the sample and shall require the establishment of a new stress relieved lot. If stress relieving or heat treatment is not performed as part of the manufacturing process, then two base plugs per five thousand base plugs produced shall be subjected to this test.

4.4.3.2 Band seating

4.4.3.2.1 Sidewall contact.-Rotating band samples obtained as specified in 4.4.3.1.3 shall be examined as described in 4.5.5.2 in order to determine compliance with the requirements of 3.4.3. Failure of any sample to comply with the applicable requirements shall reject the production represented by that sample.

4.4.3.2.2 Tightness

4.4.3.2.2.1 Nondestructive method.-Ten projectile body assemblies from each hour's production of each banding machine in operation shall be subjected to this test as specified in 4.5.5.1.1. Failure of one or more rotating bands to comply with the requirements of 3.4.4 shall reject the projectile body assemblies represented by the sample.

4.4.3.2.2.2 Destructive method.-When the nondestructive equipment is not available or unable to be utilized, two projectile body assemblies shall be selected from each eight hours of band machine operation or fraction thereof from each banding machine in operation. When tested as specified in 4.5.5, failure of any band to comply with the requirements of 3.4.4 shall reject the production represented by that sample.

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4.4.3.3 Surface preparation and application of protective finish.-The sample size and acceptance criteria for the inspection of surface preparation and the application of protective finish shall be in accordance with the applicable specifications. Failure to comply with the applicable specification and drawing requirements shall be cause for taking the corrective action recommended therein (see 4.5.6).

4.5 Test methods and procedures

4.5.1 Mechanical properties

4.5.1.1 Hardness.-After final heat treatment or stress relief, but prior to finish machining, a Rockwell hardness test shall be performed on the outside surface of the projectile body to the rear of the location of the rear rotating band seat or at any other location deemed necessary for process control. The equipment and procedures for performing this test are prescribed by ASTM Method E-18.

4.5.1.2 Tension test

4.5.1.2.1 Body.-From the samples provided by the heat treat or stress relief lots, two tension test specimens conforming to the largest standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body at midwall, in such a manner that the center of the gage length of the machined test bar will fail in line coincident with the center of the rotating band seats. The number of specimens and their location is to determine the acceptability of the mechanical properties requirements of the applicable drawing and with the approval of the procuring agency, the Government shall not be restricted from testing additional specimens at other locations to determine that the mechanical properties of the projectile conform to the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8 except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.5.1.2.2 Rotating band.-The sample provided by 4.4.3.1.3 shall have the rotating band removed and carefully flattened without hammering so as to hold additional cold work to a minimum. The specimen blank shall be machined

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so that all serration marks from the band seat are removed. A tension test specimen having a one inch gage length and a one half inch gage width with grip ends of sufficient length to accomodate the tensile machine shall be removed from the blank and tested in accordance with the procedures prescribed by ASTM Method E-8.

4.5.2 Magnetic particle inspection.-The procedures for performing this test shall be as specified in MIL-I-6868, for indirect magnetization by either the continuous method or the residual method, both circular and longitudinal magnetization shall be used. The solid content of suspension shall be maintained at 0.2 to 0.4 milliliter (ML) of solids per 100 ML of suspension when oil is used, or 0.15 to 0.35 ML of solids per 100 ML of suspension when water is used. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by Section 4 of MIL-I-6868. When performing the test for solid content of the suspension, if it is determined that it does not comply, all projectile bodies that have been tested and accepted since the last satisfactory test was performed shall be re-tested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of MIL-M-6867. The operator(s) of the equipment and the inspector(s) making the determinations shall be certified in accordance with MIL-STD-410.

4.5.3 Hydrostatic test.-The body shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. The body assembly shall be mounted in an approved test fixture in such a manner than it can be tested without constraint. In performing the test, the hydrostatic pressure will be applied gradually until the minimum required pressure is reached, afterwhich the pressure will be held for the minimum time required by the applicable drawing. The equipment shall be calibrated prior to the start of each day's operation and at intervals not over 4 hours continuous operation. It shall also be calibrated prior to restarting the operation after a layoff of 2 hours or more.

4.5.4 Air pressure test (if applicable).-This test shall be made using equipment capable of performing this test accurately and consistently. Body assemblies shall be mounted in an approved fixture in such a manner that the ogive of body assembly shall be submerged in water for a

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minimum of six inches without interference. The water in the tank shall be permitted to settle after immersion of the body assembly. Observation for leakage shall be made for the entire time the body assembly is required to be pressurized. The equipment shall be calibrated at the start of each day's operation and at intervals of 4 hours and after a stoppage of 3 hours or more. If it is found that the gage is out of calibration in such a direction that the test pressure was lower than test pressure specified on the drawing, the body assemblies tested since the last sataisfactory calibration shall be retested after the gage error has been corrected.

4.5.5 Band seating

4.5.5.1 Tightness

4.5.5.1.1 Non-destructive method.-The sample provided by 4.4.3.2.2.1 shall be tested utilizing the equipment and procedures prescribed in Appendix A to this specification.

4.5.5.1.2 Destructive method.-The sample provided in accordance with 4.4.3.2.2.2 need not be processed through the band machining line, although the band should have a light surface cut to true the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on 2 points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band; these points of measurement shall be marked on the bands and on the projectile adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat, the sum of the corresponding 2 thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.5.2 Sidewall contact.-The samples provided in accordance with 4.4.3.2.1 shall be sectioned by cutting

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through one plane normal to the longitudinal axis one-half inch forward of the forward rotating band. The cylindrical portion containing the rotating bands shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other half shall be sectioned into two parts, thus exposing four surfaces each of which reveals a view of the two bands swaged into the band seats. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effectsof saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.4.3.

4.5.6 Protective coating.-The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification.

4.5.6.1 Salt spray.-Procedures and equipment to be used in performing this test shall be as prescribed in ASTM Method B117.

4.5.7 Proving ground functional tests

4.5.7.1 Security phase.-The projectiles supplied in accordance with Table 1 and 4.4.2.6 for this phase, inert loaded to the service weight, with case impeller, projectile Burster M175 assembled into complete rounds and fired for recovery in a gun for which the projectile is standard, the tube of which will have a minimum of 25 percent life remaining, based on current gun tube condemnation criteria. A propellant charge adjusted to produce a 109 to 117 percent of the upper pressure limit of the propellant used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.6. An attempt will be made to recover at least 60 percent of the projectiles fired for examination to determine compliance with 3.6.

4.5.7.2 Performance phase.-The projectiles supplied in accordance with Table 1 and 4.4.2.6 for the phase inert loaded to the service weight with case impeller Projectile

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Burster M175 assembled into a complete round utilizing the service charge and fired alternately with projectiles from a reference lot for comparison. The projectile shall be fired in a gun for which the projectile is standard, the tube of which shall have a minimum of 25 percent life remaining based on current gun tube condemnation criteria. All rounds will be fired alternately through a canvas or heavy muslin screen placed approximately 2000 meters from the muzzle of the gun, accuracy in terms of maximum spread both horizontal and vertical will be recorded for evaluation by the technicalagency. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.6. Velocity and chamber pressure will be recorded for each round fired.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Level C.-Packing and marking shall be in accordance with MIL-STD-1169 and this specification.

5.2 Packing

5.2.1 Body assembly.-Body Assembly, Part Number 10522806 with gasket, Part Number, 8597096 taped in place shall be packed in accordance with MIL-STD-1169.

5.3 Marking

5.3.1 Body assembly.-Palletized body assemblies shall be marked in accordance with Drawing F8837835.

5.3.2 Plug, base.-Packaged base plug will be marked in accordance with paragraph 3.4 of MIL-STD-1169.

5.4 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separate and divisions between lots clearly indicated to prevent mixing in transit.

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projectile fails to comply with the applicable requirements. If any

5.5 Data cards.-Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Intended use.-The components covered by this specification are intended for use on the projectile WP 105MM-416 Metal Parts Assembly.

6.2 Ordering data.-See MIL-A-48078.

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Army-AR
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APPENDIX

Nondestructive Rotating Band Tester

10. SCOPE

10.1 The Frankford Arsenal Detroit Testing Machine Manual, shall be used when testing the sample provided by 4.4.3.2.2.1.

10.2 Instructions for Detroit Testing Machine, Model RB-1.

a. To Set Up Machine

(1) Uncrate carefully and set machine on a sturdy base of desired height.

(2) Attach foot treadle to lower back leg of bench, approximately 3" from the floor, with lag screw or bolt.

(3) Cut foot treadle rod to suit and connect treadle to operating level on right hand side of machine head. Be sure treadle does not touch floor when fully depressed.

(4) A lever can be attached to operating level for hand operation, rather than foot treadle operation as previously described.

(5) Remove hexagon eye-bolt plug from top of machine and fill to not more than half-way on oil gauge with Vacuoline medium heavy oil. Approximately 1-1/2 gallons will be required. If Vacuoline oil is not available, any good S.A.E. #30 oil may be substituted. Oil should have a viscosity of approximately 300 Saybolt. Do not use hydraulic oil.

(6) Make necessary electrical connections and operate ram up and down a few times in order to expel air from oil system. Do not run ram against the bottom head unnecessarily. The ram travel is approximately 5/8", and except as noted above, the machine should not be operated without a shell in the fixture.

(7) It will be noted that the shell fixture is supported by a coil spring having a nut adjustment. With shell in the fixture, the spring should be compressed enough to support both shell and the fixture just above lower indentor.

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(8) When the shell is in position and the upper indentor contacts the band, shell and fixture are pushed down until the bottom band rests on lower indentor, this slight clearance is adjusted by means of the socket headscrew immediately back of lower indentor.

(9) The small sliding guard bar near the top indentor is intended for protection of the indentor. The fixture and shell should be adjusted by means of the large screw and nut until the bottom of the top indentor is more than 1/4" above the top surface of the band. Ram must travel not less than 1/4" to function properly. This adjustment is made with the motor running which automatically raises ram to extreme "up" position. The large screw should be locked securely in this position by means of lock screw in base.

b. Pressure Adjustment

Insert proper indentors and place shell in fixture. Open knobbed gauge valve on top of machine. Depress foot treadle (or hand lever) firmly and hold pressure on. Check pressure gauge and, if necessary, adjust to desired pressure by means of "T" handle on top of machine.

c. Quick Calibration Check with Proving Rings

(1) Warm up machine for half an hour by turning the pump motor "on".

(2) Insert indentors provided with the proving rings or the ones used with 105mm shell (see F.A. Dwg No. D8593429).

(3) Set one of the three proving rings on the lower indentor. Any of the three rings can be used. However, in order to prevent possible damage to the rings, use P3 ring for pressures up to 6,000 lbs., P2 ring for pressures up to 8,000 lbs., and P1 ring for higher pressures. The proving ring should be set with the index mark set splitting the width of the indentor. The proving ring shall be adjusted so that approximately 1/4" exists between the proving ring and the upper indentor with the ram in retracted position.

(4) Insert a suitable dial bore gauge into the ring and zero it carefully. Apply pressure. Notice the reading on the pressure gauge and the deflection reading on the dial bore gauge. Compare

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the deflection and the pressure reading with the reading from the appropriate proving ring calibration curve supplied with the proving rings. If the readings do not compare, mark the pressure gauge accordingly.

(5) If proving rings are not available, a Morehouse calibrated proving ring or a calibrated pressure gauge can be used.

d. Test Procedure

(1) Insert proper indentors. Place shell on test fixture and adjust guide so that the rotating band of the projectile is correctly located with respect to the indentors.

(2) Step on treadle (or pull hand lever) and allow pressure to act until needle on lower or indicating dial comes to rest. Note reading.

(3) Release treadle and allow indicating needle to return to zero.

(4) Without disturbing projectile repeat Step #2 and note reading. Release treadle. The difference between the readings of Step #2 and Step #4 is the function of the gap or clearance at that diameter. Rotate shell 90° and obtain another reading.

(5) Repeat Step #2 through #4 for other shells.

e. Principle of Operation - Detroit Testing Machine - Model RB-1

(1) As illustrated, the operating valve (27-A) has been depressed and various valves, etc. are about to function.

(2) Pressure from pump enters at A and progresses through port D and oil-leads B and C.

(3) Ram descends until indentor contacts work. Further motion being arrested, pressure builds up in leads B and C and under valve (21-A) until a total of approximately 200 lbs. is applied to specimen. This preload is governed by valve (21-A) and weight (21-B).

(4) When preload point is reached, valve (21-A) is forced upward closing port D and opening port E, thus initial preload is prevented from increasing and pressure is diverted through lead F as per dotted line, to bottom of second valve G and to back of lock piston H, until a clamping load of approximately 1000 lbs. is

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applied, thus locking load piston (5) and indicating piston (11-A) securely together.

(5) It should be noted that prior to this time, indicating piston (11-A) was held up against its top by spring I and did not move down with test piston (5). Thus the indicator is now resting on the rotating band under an initial load of 200 lbs. and the test piston and indicating piston are now moving as one unit. Any further downward movement now registers on dial indicator (32) through finger (11-B).

(6) Immediately the 1,000 lbs. clamping load, referred to above, is exceeded, valve G is raised permitting flow to bottom of valve J and back through lead C to top of that piston. The load now builds up to the desired load, as determined by setting of spring on top of valve J and overflows to reservoir at K.

(7) While test piston is descending, oil on bottom end is evacuated through lead L, as indicated by solid arrows. When test is completed, operating valve (27-A) reverses, allowing oil-leads B and C to discharge into the reservoir, allowing valve (21-A) to drop, closing port E and opening port D, forcing oil back through lead L, dotted arrow, to bottom of test cylinder thus forcing piston up ready for next test.

(8) The reason for the large pipe to top of cylinder is that the test piston will descend by gravity faster than it can be pumped down. The pump serves only to apply the slow test pressure and return test piston to "up" position.

f. Balancing Bracket

Page 7 shows a 155mm shell resting on the standard "V" block supplied with the Model RB-1 Detroit Testing Machine. As can be seen from this illustration, most of the weight is resting on the "V" block. With shell of larger size, i.e., 120mm and 155mm, the greater length causes an over-hang. Those projectiles, resting on a normal bracket, have a turning point which must be counter-balanced to maintain indentor alignment. In inspecting those projectiles a balancing bracket, similar to that shown on Drawing FF-4134, should be used.

g. Immersion Heater

When this machine is to be installed in an unheated location, it should be equipped with a thermostatically

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controlled immersion heater in the oil reservoir. This will maintain the oil at the proper viscosity and temperature (approximately 100°F.).

DESTRUCTIVE METHOD OF BAND GAP MEASUREMENT

Prior to measuring, the rotating bands of the sample provided shall be given a light cut in order to true up the bands and permit accurate measurement. Before removal of the bands the diameter of the bands shall be measured on three diameters, 60 degrees apart. Points of measurement or orientation points shall be marked on the bands and on the projectile body above and below the bands in such a manner that each measurement location can be identified. Care shall be exercised in the removal of the bands in order to keep deformation to a minimum and asure that markings are not obliterated. After removal of the band, the thickness of the band shall be measured at each of the marked or orientation points. Then the diameter of the band seats at corresponding points shall be measured. To each of the measurements of diameters of the band seats the corresponding band thickness shall be added and the result subtracted from the corresponding exterior diameter of the band. The differences shall be added and the sum divided by three. This then is the average diametrical clearance between the band the band seat.

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Operating Instructions for Detroit Testing Machine Model RB-2

a. To Set Up Machine

(1) Uncrate carefully, set machine on floor and level with bolts provided at the base of the machine.

(2) Fill machine reservoir with approximately twelve (12) gallons of SAE#30 or DTE medium heavy (Socony Vacuum Oil Company) motor oil.

(3) Connect machine to 110 volts, 60 cycles, single phase electrical system.

(4) Turn heat switch to "ON". Oil will heat up in approximately one (1) hour. Turn motor switch to "ON" and allow to idle with operating level in return position for approximately five (5) minutes. Now allow piston to travel forward and return a few times to expel air from the system.

NOTE: TO PREVENT DAMAGE TO INDICATOR, ETC., DO NOT ALLOW PISTON TO MOVE FORWARD MORE THAN 3/8 OF AN INCH OUT OF CYLINDER WITHOUT SHELL OR PROVING RING BETWEEN INDENTORS.

(5) Now, move operating lever forward and then backward until white light turns on. Leave lever in back or return position. Machine is now ready for testing.

(6) The large piston must move out of cylinder 3/16 of an inch before large piston area is effective. Anything less than 3/16 indicates that small return piston is still applying its load. After large ram has moved out 3/16 from front face of cylinder the load is effective for another 5/16 of an inch of travel.

(7) Place shell on the conveyor. Open shut-off valve on high pressure gage and move operating lever forward. The lift piston will lift shell from conveyor to testing position, then small piston in rear of load piston will push large piston forward almost to shell band. Now, large piston is effective and moves forward until indentor touches band and moves shell against back indentor; then primary load builds up and piston stops forward movement momentarily. The system then locks indicating piston; load piston again moves forward with the

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indicator reading balance of travel until load is built up to load valve setting.

b. Pressure Adjustment

With shell in position, open pressure control valve behind main operating lever and also pressure gage shut-off valve. Move and hold main operating lever to the right and adjust machine to desired pressure. Turn pressure control valve clockwise for greater pressure, counter-clockwise for less pressure. After desired pressure is set, move main operating lever to return position and turn off the pressure gage shut-off valve.

c. Calibration Check

(1) A special adapter has been made to hold proving rings (FA A27413). This adapter is attached to the stationary ram and spacers are provided to align the ring with the center line of the indentor.

(2) Insert indentors provided with the proving rings.

(3) Set one of the three proving rings in the special adapter mentioned in 1. above. Any of the three rings can be used. However, in order to prevent possible damage to the rings, use P3 ring for pressures up to 6,000 lbs., P2 ring for pressures up to 8,000 lbs., and P1 ring for higher pressures. The proving ring should be set with the index mark set splitting the width of the indentor.

(4) Insert a suitable dial bore gage into the ring and zero it carefully. Apply pressure. Notice the reading on the pressure gage and the deflection reading on the dial bore gage. Compare the deflection and the pressure reading with the reading from the appropriate proving ring calibration curve supplied with the proving rings. If the readings do not compare mark the pressure gage accordingly.

(5) If proving rings are not available, a Morehouse calibrated proving ring or a calibrated pressure gage can be used.

d. Test Cycle

(1) Place shell on roller conveyor and adjust shell stop so that the rotating band of the projectile is correctly located with respect to the indentor.

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(2) Push shell to shell stop and move lift piston lever to the right. Shell is now ready for testing.

(3) Move main operating lever to the right and allow load to act until needle on indicating dial comes to rest. Record reading.

(4) Move main operating lever back to the original position and allow indicating needle to return to zero.

(5) Repeat Steps #3 and #4.

(6) The difference between the readings of Step #3 and Step #5 is the function of the gap or clearance at that position.

(7) Index shell 90° and repeat Steps #3 through #6.

(8) Move lift piston lever to the left and remove shell.

e. Test Stroke - See Dwg No F8595564

Idling Position: With operating valve and lift valve in position shown, flow is to reservoir. Closing work stop moves lift valve lever to position shown in phantom which allows lift valve to close and renders machine operative.

Phase 1: Oil under pressure flows from pump through line 1, through operating valve assembly to dimple in test piston (this helps to counter-balance weight of piston), and to the lift valve assembly.

Phase 2: Flow continues past ball check through line 2 to the lift piston which rises with shell to test position (ball check keeps lift piston up until lift valve lever is depressed). When lift piston is up, pressure builds to raise lift regulating valve.

Phase 3: Oil through line 3 moves through load valve assembly to top of return piston (small area of return piston permits fast forward action). Oil then moves through return cyclinder to test cylinder and acts upon test piston. Flow continues through line 3, through operating valve assembly to bottom of preload valve. When test piston has reached the preload setting pressure builds up and raises preload valve stopping forward motion of test piston.

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Phase 4: Oil continues through load valve assembly, through line 4 to lock valve assembly. Lock valve assembly halts flow of oil momentarily and permits oil to engage lock piston with full impact insuring locking of indicator mechanism so that further movement of test piston can be measured. The two dimples on test piston connected to indicator locking mechanism equalize the load caused by the locking action. After indicator assembly is locked raising pressure raises regulating valve.

Phase 5: Flow continues through test regulating valve, through line 5, through operating valve assembly to back of test piston. Pressure builds up to full test load as determined by setting on test load valve assembly.

Phase 6: Exceeding test load raises test load valve which discharges to reservoir.

f. Return Stroke - F8595565

Phase 1: Operating valve is moved to position shown. Oil under pressure moves from pump through operating valve, through line 1 to bottom of return piston forcing it to retract test piston.

Phase 2: Oil at the head of test piston escapes through line 2 through operating valve to reservoir.

Phase 3: Oil at head of return piston escapes through line 3, preload valve, and lift regulating valve to reservoir.

Phase 4: Opening work stop depresses lift valve. Lift piston lowers due to gravity and oil escapes through line 4 and lift valve to reservoir.

Phase 5: Oil under valve piston in lock valve assembly escapes through line 5 and preload valve to reservoir.

Phase 6: Oil under lift regulating valve and ball check escapes through line 6 and operating valve to reservoir.

Phase 7: Oil under preload valve escapes through line 7 and through operating valve to reservoir.

Phase 8: Oil under test load valve escapes through line 8 and operating valve to reservoir.

The purpose for the large pipe and swing check valve is to permit free flow of oil to and from test cylinder on both forward and return strokes.

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